Computer Graphics Society

Graphics Task

Hello sophomores! Welcome to the task round of the Computer Graphics Society, Graphics team. This challenge is designed to test your creativity, technical skills, familiarity with 3D modeling and animation using Blender and to bring your imagination to life. Whether you are new to Blender or have some experience, this task will help you build a strong foundation for more advanced projects.

Do at least one task

More is welcome, but quality over quantity.

TASK DEADLINE: 27 JUNE 2026

Join our WhatsApp Group for Selections

<Group link>

TASK 1: Looping Animation

Objective:

Create a detailed and visually appealing looping animation of your imagination in Blender. This task will test your skills in **animation and motion graphics**.

Task Requirements:

- **Design and Modelling**
  Model and texture the background, environment and the main objects for the animation loop. Try to create an aesthetically pleasing setup. You can use external textures and HDRIs. External models for the background must adhere to the 60-40 rule i.e. 60% should be self-made and 40% can be external assets. The foreground object should be fully self-made.

- **Video and Animation**
  The video must be in 4:3 aspect ratio. Animation must be done at 24 fps, and the total time must lie between 5-8 seconds. The starting and ending frame must be the exact same and the loop must be smooth.

- **Optional Enhancements**
  Any relevant moving objects in the background carry extra points. For example, a cloth swaying or a tap dripping.

- **Examples and references**

  Blender Satisfying Looping Animation Compilation
  5 Ways to Make Looping Animations in Blender 3.4 (Blender Tutorial)

TASK 2: Mechanical Rigging

Objective:

Create a detailed mechanical model with working joints, and a complete armature using Blender.

Task Requirements:

- **Model**
  Try to model as many parts of the model as possible, especially the joints. Smaller parts like screws, nuts and bolts can be imported as assets. Try to strike a balance between lower polycount and higher complexity.

- **Idea**
  The more complex the model and the joint movements, the better. However, we don't expect full accuracy. For example, parts can move in and out of places with no space, but make it believable. This is not an engineering challenge.

- **Optional Enhancements**
  You may add additional features to enrich the scene such as animation, sound effects, background elements and more.

- **Presentation**
  Submit 2-3 renders of the model in different states (e.g. open or closed, expanded or contracted etc.). Also render a video of the model transitioning between its states.

- **Examples and References**
  How to create a mechanical rig - short Blender tutorial
  Rigging an Aircraft Landing Gear - A Blender tutorial - Beginner Guide

TASK 3: Simulation

Objective:

Create a detailed and visually appealing 3D render of a **realistic phenomenon.** This task will test your skills in **simulation**.

Task Requirements:

- **Design and Modelling**
  Model and texture the background, environment and the main objects for the simulation. Try to create an aesthetically pleasing setup. You can use external textures and HDRIs. External models for the background must adhere to the 60-40 rule i.e. 60% should be self-made and 40% can be external assets. The foreground object should be fully self-made.

- **Simulation and Addons**
  Use Blender's Physics system to create a smooth realistic simulation that interacts with relevant environment objects. Complexity is rewarded but inaccurate effects or glitches will be penalized. You can use any external addons.

- **Video and Animation**
  The video must be in 4:3 aspect ratio. Animation must be done at 24 fps, and the total time must lie between 5-10 seconds. The starting and ending frame must be the exact same and the loop must be smooth.

- **Examples and References**
  [The Most Intense Blender 3D Physics Simulation [Compilation] ☐ | 0% → 1000%]
  [The Most Satisfying Blender 3D Physics Simulation | 8 Minutes of Non-Stop Satisfaction]

IMPORTANT:

- **Refrain from copying directly from any tutorial or reference image/video for any of the above tasks. Any use of AI, (except for ideation and creating reference images) is prohibited. These will be grounds for instant disqualification from the selection process.**

Submission Requirements:

To submit your work successfully, follow these detailed steps. We've simplified the process for you:

1. **Prepare Your Folder:**
   - Your Blender project file (.blend)
   - Some rendered images of your project in different camera angles.
   - One video file of the final animation
   - 3-4 clear screenshots of the scene in the viewport in wireframe and solid views.
   - Name of the folder: **CGS_[Rollno]_[YourFirstName]_Graphics**
2. **Upload the folder**
   - Open Google Drive and Upload the Folder.
   - Copy the link for the folder after changing the access to "Anyone with link"
   - Paste the link in the Google Form which we will provide near the submission deadline

Some tutorials to help you in the journey towards Graphics :)

- **Blender Basics** : [CrossMind studio absolute beginner blender series] **(GO THROUGH AT LEAST DAY 1 TO DAY 3 TO BEGIN WITH MODELING)**
- **Blender speed issue** : EVERY way to SPEED up Cycles! Up to 1000% - Blender 3D - YouTube
- **Animation in blender** : Animation for Beginners! (Blender Tutorial) - YouTube
- **Sculpting in blender :** Sculpting for beginners
- **Donut Series By BlenderGuru :** **Beginner donut series**

Remember, these tasks are meant to be enjoyable and provide an opportunity to showcase your skills in 3D modeling and creativity. Feel free to reach out if you have any questions, and have fun bringing your ideas to life in Blender!